

Research Statement

I am a macroeconomist with interests in labor markets and search. At the core of the study of the aggregate labor market is a desire to understand the causes of underutilization of resources in the aggregate economy, made especially tangible in the unemployment of workers. One of the biggest advances in our understanding of the labor market has come through the use of search theory, capturing the intuitive notion that the matching of workers and firms is costly in time and effort due to the different preferences, skills, and information possessed by workers and firms. My research focuses on the use of micro level data and computational modeling to study the interaction of search frictions and unemployment with various aspects of the macroeconomy. I address the fact that frictions in the labor market carry over into other dimensions of firms' and workers' decisions, such as a firm's incentive to utilize its current labor force, workers' participation in the labor market, and the decision to acquire or discharge debt.

Part-time Employment and Firm-level Labor Demand over the Business Cycle [Job Market Paper]

In addition to unemployment, workers may be underutilized through being employed part-time when they would desire full-time work. The population at work part-time for economic reasons (*PTE*) is countercyclical and fluctuates with the business cycle with similar volatility to the unemployment rate. The popular narrative of the involuntary part-time worker involves the recently unemployed who, for lack of better alternatives, has settled on part-time work. However, I show using data from the Current population survey and Survey of Income and Program Participants that the cyclical fluctuation in the population of part-time employed for economic reasons comes from the movement of workers between full-time employment and part-time rather than between part-time and unemployment. Not only is the fluctuation in *PTE* from within-employment, but it is largely due to within-job changes in hours.

Since fluctuations in *PTE* are driven by changes in the hours demanded of existing employees rather than changes in workers' job search or acceptance behavior, I model part-time work focusing on a firm's decision to hire, fire, or partially utilize its labor force in the face of firm-level and aggregate shocks. This requires a search model where firms are heterogeneous in size and productivity. The model produces part-time labor movements which are consistent with the evidence from worker flows in the CPS and match the relative volatility of unemployment and part-time for economic reasons over the business cycle. Part-time labor utilization by firms increases the volatility in vacancies and unemployment in the model relative to the model with only an extensive margin. Due to heterogeneity in firm size and productivity, the model has implications for the size and age distribution of firms and the growth rates of hours and employment.

The Role of Firm Size and Age on Employment and Hours Adjustments [Work in progress]

The role of firm characteristics such as size and age in the tradeoffs between hours and employment are highlighted in my model, but data on these adjustments at the firm-level is lacking. In ongoing work, I examine the cyclical dynamics of employment and hours across the distribution of firms. I utilize firm-level longitudinal micro data from the Longitudinal Business Database, Census of Manufactures, Annual Survey of Manufactures, and Quarterly Plant Capacity Utilization Survey at the U.S Census Bureau's Research Data Centers to measure the comovements and cyclical differentials in employment and hours adjustments by firm size, age, and productivity.

Matching Labor Flows in Search Models with Labor Force Participation [Working Paper]

Introducing a participation margin in a search model requires the use of large idiosyncratic shocks to workers' participation incentives to match the labor flows in the data. If we measure the participation transitions of workers outside of employment where search decisions are observable and apply this same transition process to employed workers, any search model will overstate the transition of workers out of employment to nonparticipation. Allowing the participation transition of workers to depend on their employment state fixes these flows, but this transition process is unobservable for employed workers. Taking advantage of the longer panel of the 1996 Survey of Income and Program Participants, I estimate the markov process for participation transitions of employed workers using their observed search behavior before and after an employment spell. The difference in the transition process measured for employed and nonemployed workers is consistent with an interpretation of attachment to the labor force. I build a directed search model with a labor force participation margin subject to employment-dependent shocks and show that it can match the labor market flows in US data.

Student Loans and Labor Market Outcomes [Working Paper] (with Chander S. Kochar)

The student loan market is the second largest source of household debt in the United States, with \$1.2 trillion in outstanding debt. Unlike other sources of unsecured credit, student loans cannot be discharged in bankruptcy. Using data on college graduates from the 1993/03 Baccalaureate and Beyond Longitudinal Study, we first identify that student loan debt has a significant negative effect on students' earnings after graduation. We show that the inability to discharge debt in bankruptcy is critical to produce this result within a simple search theoretic framework. We propose a richer model with student loan debt and a delinquency/default decision to study the effects of recent changes to student loan policies on the labor market and delinquency outcomes of college graduates.